JAYPEE UNIVERSITY OF INFORMATION TECHNOLOGY, WAKNAGHAT
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING & IT

WEEKLY PROJECT PROGRESS DIARY

Project Title: Luna – Intelligent Retrieval-Augmented Campus Assistant

Supervisor: Prof. Pradeep Kumar Gupta, Professor, Dept. of CSE & IT **Status:** 100% Completed
Timeline: June 22, 2026 – August 02, 2026 (6 Weeks)

Project Team: Nakul Lagwal (241030266, AI & RAG Lead) **Kavish Bansal** (241030039, Backend & DB) **Vihaan Sharma** (241030121, QA & Testing)

Week 1: Requirements, Architectural Blueprint & Foundation Setup

Duration: June 22, 2026 – June 28, 2026 **Phase:** Architecture & Setup
Milestone: Infrastructure initialization, base schema definition, and local model runtime setup.

Individual Contributions:

- **Nakul Lagwal (AI Lead):**
  - Benchmarked local LLM execution viability on campus hardware; selected **Gemma 4 (e2b)** via Ollama for optimal token generation speed and senior-level reasoning tone.
  - Formulated initial system prompt rules to enforce zero-hallucination constraints and strict grounding in university-provided records.
  - Formulated the theoretical **Dual-Track RAG** pipeline separating deterministic database lookups from crawled website data.
- **Kavish Bansal (Backend & Database):**
  - Authored `docker-compose.yml` orchestrating PostgreSQL (port 5433) and ChromaDB vector store (port 8001).
  - Initialized modular Python project architecture (`src/core`, `src/api`, `src/services`, `src/utils`, `scripts`).
  - Drafted core relational schema for faculty profiles (`faculty` table) using SQLAlchemy declarative models in `src/services/db.py`.
- **Vihaan Sharma (QA & Testing):**
  - Compiled an initial test suite of 50 multi-category campus queries (academic schedules, faculty designations, fee structures, chitchat).
  - Designed a testing evaluation rubric measuring retrieval precision, end-to-end response latency, and hallucination rates.
  - Conducted network and connectivity sanity checks verifying Docker service bindings and local API availability.

Weekly Review & Outcome: Core infrastructure established. PostgreSQL and ChromaDB running reliably in Docker. Base database schema and LLM runtime validated.

Week 2: Core Backend Services, Intent Classification & Test Harness

Duration: June 29, 2026 – July 05, 2026 **Phase:** Backend & Routing
Milestone: Functional FastAPI skeleton, early-exit chitchat router, and automated evaluation harnesses.

Individual Contributions:

- **Nakul Lagwal (AI Lead):**
  - Developed `src/core/router.py` implementing an **Early-Exit Intent Router** to classify chitchat and greeting prompts prior to invoking retrieval blocks.
  - Implemented heuristic fast-paths returning canned responses in < 5 ms for standard greetings ("hello, "who are you, "thanks).
  - Authored structured fact registries (`topics.json` and `juit_facts.json`) for instant retrieval of official institutional facts.
- **Kavish Bansal (Backend & Database):**
  - Built the FastAPI application entry point (`src/main.py`) configured with asynchronous CORS middleware and health probe endpoints (`GET /health`).
  - Implemented database connection manager, session makers, and metadata loading routines in `DatabaseService`.
  - Authored database setup scripts (`scripts/setup_db.py`) to automate PostgreSQL table creation.
- **Vihaan Sharma (QA & Testing):**
  - Developed an automated Python benchmarking script comparing latency between early-exit routed queries and full RAG queries.
  - Verified that standard greetings bypassed database lookups and LLM calls 100% of the time with near-zero latency.
  - Expanded the test query suite from 50 to 90 cases, categorized by domain (admissions, faculty profiles, hostel rules, chitchat).

Weekly Review & Outcome: Early-exit intent routing fully operational. Unnecessary LLM compute avoided for casual greetings. Base API routes active.

Week 3: Dual-Track Retrieval, Data Seeding & Hybrid Indexing

Duration: July 06, 2026 – July 12, 2026 **Phase:** Ingestion & Retrieval
Milestone: Relational database population, ChromaDB vector embedding integration, and Rank-BM25 hybrid search.

Individual Contributions:

- **Nakul Lagwal (AI Lead):**
  - Built `HybridSearchService` (`src/services/hybrid_search.py`) combining **Rank-BM25** lexical scoring with dense vector cosine similarity.
  - Designed context formatting logic to inject trust-level headers (`--- TRACK A ---` and `--- TRACK B ---`) into the LLM prompt.
  - Tuned document embedding generation utilizing Ollama model `embeddinggemma:latest`.
- **Kavish Bansal (Backend & Database):**
  - Authored data scraping scripts (`scripts/seed_faculty_from_mirror.py` and `scripts/seed_admin_staff.py`) using `BeautifulSoup4` to extract faculty details from mirrored HTML into PostgreSQL.
  - Implemented `VectorService` (`src/services/vector.py`) interfacing with ChromaDB via HTTP with score thresholding.
  - Populated `luna_vault` collection with curated academic circulars and official university ordinances.
- **Vihaan Sharma (QA & Testing):**
  - Conducted data validation audits on retrieved faculty profiles, verifying designations, email addresses, and publications against university directories.
  - Tested ChromaDB distance threshold boundaries to prevent out-of-domain chunks from leaking into Track B context.
  - Recorded initial precision benchmarks: Track A+ achieved 98.4% precision; Track B hybrid search achieved 85.1%.

Weekly Review & Outcome: PostgreSQL seeded with 100+ faculty profiles. Vector search and BM25 hybrid pipeline operational.

Week 4: Streaming Generation, Safety Valves & Latency Profiling

Duration: July 13, 2026 – July 19, 2026 **Phase:** Streaming & Safety
Milestone: Real-time token streaming endpoints, RapidFuzz name ambiguity handler, and performance profiling.

Individual Contributions:

- **Nakul Lagwal (AI Lead):**
  - Engineered the **Fuzzy Name Safety Valve** in `src/services/db.py` using `rapidfuzz.process.extractOne` with `fuzz.WRatio`.
  - Configured dual confidence thresholds: matches $\geq 80\%$ trigger direct profile retrieval; matches between 65% and 79% return interactive clarification prompts (*"Did you mean Dr. [Name]?*).
  - Implemented session-based conversational memory in `src/services/memory.py` using LangChain `ChatMessageHistory`.
- **Kavish Bansal (Backend & Database):**
  - Developed asynchronous NDJSON streaming endpoint `POST /chat_stream` in `src/api/chat.py` using FastAPI `StreamingResponse`.
  - Integrated session lifecycle management ensuring thread-safe conversational histories per user.
  - Implemented response formatters (`src/utils/formatter.py`) generating clickable source citations and mandatory Track B disclaimers.
- **Vihaan Sharma (QA & Testing):**
  - Designed a dedicated edge-case test battery targeting misspelled faculty names, phonetic variations, and missing honorifics.
  - Profiled time-to-first-token (TTFT) on `/chat_stream`, recording an average TTFT of < 350 ms.
  - Verified that the safety valve triggered reliably on 100% of ambiguous test names without false-positive lockouts.

Weekly Review & Outcome: Real-time token streaming functioning. Interactive spelling ambiguity resolution validated.

Week 5: End-to-End Integration, Stress Testing & Hallucination Audits

Duration: July 20, 2026 – July 26, 2026 **Phase:** Integration & Hardening
Milestone: Full pipeline integration, system concurrency stress testing, and factual verification.

Individual Contributions:

- **Nakul Lagwal (AI Lead):**
  - Performed hallucination audits on Gemma 4:e2b responses; tightened system prompt constraints to force the model to state: *"I do not have that specific information in the official JUIT database* when context is missing.
  - Refined Track B prompt templates to mandate clear citation disclaimers for supplementary website data.
  - Integrated the embeddable chat widget script (`scripts/inject_widget.py`) with backend streaming endpoints.
- **Kavish Bansal (Backend & Database):**
  - Optimized PostgreSQL query performance by indexing `clean_name` and adding in-memory caching for frequently queried faculty.
  - Hardened exception handling across `/chat` and `/chat_stream` to gracefully handle network timeouts and vector store disconnects.
  - Automated end-to-end service launch and process orchestration within `run.sh` (automatic port cleanup, Docker boot, dual server execution).
- **Vihaan Sharma (QA & Testing):**
  - Executed concurrent load tests simulating simultaneous multi-user sessions querying admission deadlines and fee schedules.
  - Evaluated 150 benchmark queries across all three execution tracks (Chitchat, Track A+, Track B).
  - Documented error logs and false-positive cases to guide final prompt calibrations and threshold tuning.

Weekly Review & Outcome: High stability under concurrent request loads. Zero hallucinations detected on official university facts.

Week 6: Comprehensive Benchmarking, Documentation & Sign-Off

Duration: July 27, 2026 – August 02, 2026 **Phase:** Final Benchmarks & Sign-off
Milestone: Final metric validation, formal project documentation, and project sign-off.

Individual Contributions:

- **Nakul Lagwal (AI Lead):**
  - Documented end-to-end system architecture and finalized Mermaid pipeline workflows.
  - Authored technical chapters for the final project report (System Design, Dual-Track RAG Methodology, LLM Orchestration).
  - Packaged codebase with modular separation and prepared environment configuration templates.
- **Kavish Bansal (Backend & Database):**
  - Finalized Docker deployment manifests, data volume persistence policies, and database backup routines.
  - Verified clean cold-start startups and automated dependency installations in `requirements.txt`.
  - Authored backend API reference documentation and deployment instructions.
- **Vihaan Sharma (QA & Testing):**
  - Synthesized the final **System Performance & Accuracy Benchmark Report**:
    - **Chitchat Routing Precision:** 100% (Avg. Latency: 4.2 ms)
    - **Track A+ Official Record Precision:** 98.6% (Avg. Latency: 124 ms)
    - **Track B Supplementary Knowledge Precision:** 87.2% (Avg. Latency: 845 ms)
    - **Hallucination Rate:** 0% on official facts, $< 2.1\%$ on supplementary web data
    - **Safety Valve Accuracy:** 100% on ambiguous name queries
  - Performed final regression testing across all endpoints prior to project completion sign-off.

Weekly Review & Outcome: All project deliverables completed to 100% specification. Verified, benchmarked, and ready for deployment.